

Key Factors Influencing Sales Forecasting in the Food and Beverage Industry: An Integrated Approach

Nguyen Thi Ngoc Nhung
School of Management Technology
Sirindhorn International Institute of Technology,
Thammasat University
Pathum Thani, Thailand
m6622040829@g.siiit.tu.ac.th

Sarinpas Kornwattanakit
School of Management Technology
Sirindhorn International Institute of Technology,
Thammasat University
Pathum Thani, Thailand
sarinpas@siit.tu.ac.th

Accurate forecasting is crucial in Vietnam's rapidly growing and competitive food and beverage (F&B) industry. This study examines both internal factors, such as pricing, promotions, inventory levels, advertising, and e-commerce, as well as external factors, including macroeconomic conditions, government policies, significant events, and market fluctuations, that impact forecast accuracy. The findings suggest that demand forecasting in the F&B industry is highly sensitive to changes in consumer behavior, digital transformation, and external shocks, including the COVID-19 pandemic. The interaction between e-commerce growth, consumer sentiment, and market fluctuations causes rapid demand shifts that traditional models often fail to capture. Additionally, holidays, promotions, and price changes can lead to short-term demand spikes, underscoring the need for flexible, event-driven forecasting methods. This study highlights that combining quantitative data with internal and external variables enhances forecasting precision. An integrated and adaptive approach enables businesses to make informed decisions in inventory management, supply chain operations, and customer strategies.

Keywords: Factor, Forecasting, F&B industry, Market influencing, Consumer preference

I. INTRODUCTION

Vietnam's food and beverage (F&B) industry plays a crucial role in the national economy, contributing 17.2% to the GDP [1]. In 2023, industry revenue increased by 10.87% to reach USD 23.6 billion. This growth is expected to continue, with Euromonitor projecting a compound annual growth rate (CAGR) of 10.25% between 2023 and 2027, resulting in a market value of \$ 34.9 billion by 2027 [2]. Additionally, the market is forecasted to expand by 10.92% in 2024, creating both opportunities and intensifying competition for domestic and international businesses [3]. These trends underscore the importance of understanding how key factors influence market behavior, as external disruptions and evolving consumer preferences can significantly impact sales performance.

To provide empirical evidence of these dynamics, infographic representations are utilized in this study to illustrate the relationship between influencing factors and market behavior in Vietnam's F&B sector. For example, according to the annual report, the chart depicting Vinamilk's revenue trends over recent years highlights how fluctuations in consumer demand, economic volatility, and supply chain disruptions directly affect business outcomes. Furthermore, comparing Vinamilk's revenue growth with that of the overall F&B industry reveals the alignment of company-specific trends with broader market dynamics. These visual insights

reinforce the credibility of the integrated approach proposed in this research, demonstrating its capacity to capture the complex and dynamic nature of sales forecasting in Vietnam's F&B industry.



Figure 1. Revenue Trends of Vinamilk and Vietnam's F&B Industry (2014–2024)

Despite this promising forecast, demand forecasting remains a significant challenge for the food and beverage (F&B) industry. Cost fluctuations, evolving consumer preferences, and market uncertainties make it challenging for businesses to manage their supply chains efficiently and maintain profitability. The global beverage market is projected to reach 320.5 billion USD by 2025 [4], but inflation and supply chain disruptions pose significant risks to this growth [5].

Several key factors influence the accuracy of demand forecasting in the F&B sector. Macroeconomic indicators such as GDP growth, unemployment rates, consumer confidence, and government policy have a direct impact on consumption patterns [6]. Seasonal factors also play a crucial role—for instance, demand for mooncakes spikes during the Mid-Autumn Festival [7]. Additionally, consumer preferences shift with the weather, favoring light foods and cold beverages in summer and heartier meals in winter. These fluctuations pose challenges in inventory management, especially for perishable goods. Labor demand also fluctuates, requiring businesses to balance workforce shortages in low-demand periods with increased hiring during peak seasons [8].

Consumer behavior changes further complicate forecasting. The increasing demand for organic and eco-friendly products, particularly among younger consumers, is reshaping purchasing habits, with many opting to reduce their consumption of non-essential items, such as coffee [9]. The rise of digital transformation is also redefining market dynamics, with 70% of Vietnamese consumers having purchased food online, marking a trend that is expected to continue growing [10]. Post-pandemic concerns regarding health, hygiene, and food safety have become more prominent [11], reinforced by regulatory frameworks such as Decree

15/2018/ND-CP, which enforces food origin transparency and safety standards [12].

To navigate these challenges, businesses must closely integrate demand forecasting with inventory management to maintain optimal stock levels and avoid shortages or overstocking. Vietnam's F&B market is also uniquely influenced by cultural and festival-driven demand spikes, such as Tet and Mid-Autumn [13,14]. These short-term fluctuations differ from global trends, necessitating that businesses adopt precise and adaptable forecasting methods to capitalize on market opportunities while minimizing risks.

Based on this context, this study aims to identify the key factors influencing demand forecasting accuracy in Vietnam's food and beverage (F&B) industry. By identifying major influencing factors and evaluating their impact, this research aims to provide insights for enhancing forecast reliability, thereby supporting businesses in making more informed and effective supply chain decisions.

II. LITERATURE REVIEW

Accurate demand forecasting is crucial for businesses, enabling them to make informed decisions and adapt to market dynamics [15]. Instead of relying solely on historical trends, incorporating dynamic variables into forecasting models enhances precision, enabling businesses to adjust their strategies proactively in response to market fluctuations. By integrating historical data with real-time insights, companies can refine forecasts, optimize resource allocation, and mitigate financial risks [16]. Several key factors influence the accuracy of demand forecasting in the food and beverage (F&B) industry.

1. Price

Price is the monetary value assigned to a product, influenced by production costs, target market segments, and consumers' purchasing power, among other factors [17]. Economic principles suggest an inverse relationship between price and demand. When the price of a commodity decreases, demand generally increases, and vice versa, assuming other factors remain constant [19]. In the fast-food industry, pricing plays a crucial role in driving sales performance and cost control, directly impacting business profitability and brand positioning [18].

2. E-commerce

E-commerce refers to the buying and selling of goods and services over the internet, involving financial transactions and data exchange [20]. The retail industry has increasingly shifted toward online platforms, benefiting from reduced physical space requirements for product display and customer interaction [21]. The COVID-19 pandemic accelerated this trend, with research primarily focusing on online retail platforms [22]. Consequently, the rise of e-commerce necessitates integrating digital platform data into demand forecasting models to enhance accuracy.

3. Advertisement

Advertising involves the strategic dissemination of information through paid media channels, including television, newspapers, radio, digital platforms, and promotional materials [23]. Each advertising medium has distinct advantages and limitations, requiring businesses to evaluate their effectiveness carefully before implementation

[24]. The primary goal of advertising is to maximize consumer reach, ensuring that marketing efforts effectively influence purchasing decisions [25].

4. Consumer Composite Sentiment Index

Consumer psychology plays a vital role in demand forecasting, especially during economic downturns and crises. The Consumer Composite Sentiment Index (CCSI) measures consumers' willingness to spend and has been used to assess economic conditions. For example, during the COVID-19 pandemic, consumer willingness to purchase in South Korea dropped to levels lower than those observed during the 2008 financial crisis [26]. Incorporating CCSI data into forecasting models helps businesses predict consumer behavior during periods of economic uncertainty.

5. Events

Holidays and festivals have a significant impact on consumer demand, with variations influenced by location, religion, demographics, and cultural practices. Retailers situated in tourist areas or near national borders experience heightened demand fluctuations during festive seasons [28]. Consequently, seasonal forecasting must align with the event calendar and emerging consumer trends to ensure accurate inventory planning and management.

6. Physical Environment

The physical environment plays a crucial role in demand forecasting across multiple industries, including energy, tourism, and supply chain management. In logistics, adverse weather conditions can disrupt supply chains and alter consumer purchasing behavior. To mitigate these challenges, judgmental forecasting, incorporating expert insights on environmental impacts, is often used to refine predictions [31]. Additionally, integrating environmental data into hierarchical forecasting models enhances accuracy by accounting for spatial and temporal variations [32].

7. Promotion

Sales promotions are time-bound marketing activities designed to elicit direct consumer responses by offering additional benefits [33]. These strategies are integral to the promotional mix, aiding companies in market entry and customer retention [34]. Effective sales promotions strengthen customer engagement and brand loyalty. Marketing communication strategies, including advertising, public relations, and direct marketing, also influence consumer purchase intentions [35]. Tanure [36] categorizes promotional incentives into four dimensions: (1) monetary vs. non-monetary benefits, (2) immediate vs. delayed incentives, (3) product-related vs. price-related offers, and (4) economic vs. psychological promotions.

8. Macroeconomic

Macroeconomic indicators, such as inflation, have a significant impact on consumer purchasing power. Inflation erodes the value of money over time, reducing the purchasing power of a unit of currency [17]. As prices increase, consumers may adjust their spending patterns, which in turn affects demand for food and beverage (F&B) products. According to VP [37], inflation-induced price hikes reduce real income, limiting overall consumption and economic activity.

9. Historical sales

Historical sales data serves as a fundamental input for demand forecasting, offering insights into past consumer behavior and market trends. Proper data collection and preprocessing are essential to ensuring forecast accuracy. Han emphasizes the importance of reliable data sources and preprocessing techniques, such as data cleaning and normalization, in building effective sales forecasting models [38]. In industries such as pharmaceuticals, structured historical data plays a pivotal role in enhancing predictive accuracy.

10. Government policy

Government policy is defined as regulations, incentives, and interventions aimed at influencing the market [41]. The Vietnamese government's issuance of Decree No. 168/2024/NĐ-CP, which enforces stricter penalties for blood alcohol violations, has directly affected on-premise beer consumption behavior. Since January 2025, many restaurants and pubs have reported a noticeable decline in beer sales, especially during evening hours. Without timely adjustments to forecasting models to reflect this policy-driven shift in consumer behavior, businesses risk overstocking or revenue loss. Therefore, government regulation should be treated as a critical exogenous variable in demand forecasting for alcoholic beverages [42].

11. Inventory level

Effective inventory management is vital for business sustainability and profitability. According to Maslim [43], maintaining optimal stock levels boosts revenue and consumer confidence. Stock shortages lead to lost sales and dissatisfied customers, while overstocking ties up capital and increases storage costs. To optimize inventory, businesses must monitor demand trends and adjust stock levels accordingly. Kalaoglu [44] highlights that retailers should analyze market fluctuations to prevent stockouts and overstocking, ensuring efficient capital turnover.

12. Market fluctuations

Market fluctuations impact economic stability, especially in resource-dependent industries. Hevia defines a crisis as a GDP drop of at least 9%, with recovery marked by a 50% rebound [45]. South Africa's gold industry, once a global leader, has experienced declining production, which has negatively impacted GDP growth [46]. Accurate revenue forecasting is essential for estimating mining output and its economic impact [47]. Businesses must integrate scenario planning into forecasting models to adapt to sudden market shifts and ensure accuracy.

III. METHODOLOGY

A. Key factors influencing the forecast

Forecasting is crucial in the food and beverage (F&B) industry, enabling businesses to optimize their supply chains, mitigate risks, and align production with consumer demand [48]. Accurate predictions help prevent overstocking or shortages, ensuring product availability and enhancing customer satisfaction [49]. Additionally, effective forecasting enables businesses to adapt to shifting consumer trends, economic fluctuations, and unforeseen disruptions, such as pandemics, thereby fostering long-term competitiveness and sustainability.

A range of factors influences demand forecasting in the F&B industry, broadly categorized into internal and external

factors (Figure 2). Internal factors, such as pricing, e-commerce activity, promotions, advertising, historical sales, and inventory levels, are within a company's control and can be adjusted to meet short-term sales targets. In contrast, external factors, including macroeconomic conditions, consumer sentiment, government policies, market fluctuations, and unforeseen events, are less controllable but significantly impact demand [50]. Understanding the interaction between these factors is crucial for improving forecast accuracy and ensuring effective decision-making.

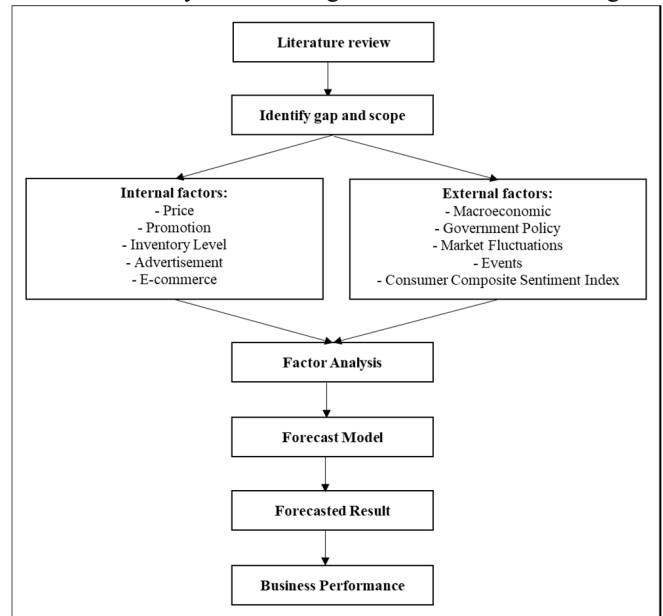


Figure 2. Conceptual framework of the forecasting model

B. Integrated approach

According to Figure 2, the proposed integrated approach combines internal historical data (e.g., sales volume, pricing, promotions) with external, real-time inputs (e.g., consumer sentiment indices, macroeconomic indicators, and weather conditions). The integrated forecasting model is designed to enhance accuracy by combining internal operational data with external contextual information. Traditional forecasting models in the F&B industry primarily rely on historical sales data, which often fails to capture sudden market fluctuations, behavioral changes, and external shocks. To overcome these limitations, the proposed approach introduces a multi-layered integration logic that incorporates diverse data types into a unified predictive framework.

1. Internal data

Internal data, such as historical sales, price, inventory levels, promotion schedules, and e-commerce performance metrics, provides the foundational structure for demand forecasting. This data is structured, time-stamped, and recorded in enterprise resource planning (ERP) or internal systems. These variables are used to identify seasonal patterns, sales trends, and the impact of past marketing interventions.

2. External data

External data includes variables such as consumer sentiment (from social media or surveys), macroeconomic indicators (e.g., inflation rate, GDP growth), government policy changes, weather conditions, and public holidays or special events. These factors are more volatile and

unstructured but critically influence consumer behavior and demand fluctuations.

To integrate these factors, an integrated approach is required. The next section will propose the steps of how to collect and integrate some approaches, such as EFA or ARIMA, to analyze the F&B sales in Vietnam.

C. Methodological approach.

The methodological approach of this study is presented in Table 1, which details each step from data collection to forecasting, along with corresponding descriptions and tools/approaches applied at each stage.

Table 1. Methodological Framework

Step	Description	Tools/Approaches
Data collection	Collect internal variables (sales, price, inventory level, advertisement, e-commerce) and external variables (macroeconomic indicators, government policies, market fluctuations, consumer sentiment index, events, weather). Data sources include financial reports, previous studies, and reliable online databases.	Literature data, Company reports, online sources
Data screening	Perform data cleaning to remove outliers, missing, or erroneous values. Apply Exploratory Factor Analysis (EFA) to identify latent variable groups, reduce the number of input variables, and focus on key factors affecting demand. This simplifies the forecasting model and enhances accuracy.	EFA
Model development	Use the screened dataset to develop a demand forecasting model. Apply the ARIMA (AutoRegressive Integrated Moving Average) model to analyze time series data, identifying trends, seasonality, and residual patterns to produce precise forecasts.	ARIMA
Forecasting	Generate demand forecasts using the developed ARIMA model. Visualize the forecast results with charts and dashboards to support inventory, production, and distribution decisions. Compare forecasted and actual results to assess forecast accuracy.	Excel, dashboard

IV. ANALYSIS AND DISCUSSION

A. The interaction of Forecasting factors

These factors are highly interconnected, influencing each other throughout the forecasting process. To improve accuracy, businesses must move beyond relying solely on historical sales data and incorporate real-time market insights, event-driven data, and qualitative inputs, such as customer feedback. This is particularly challenging in the F&B industry, where perishable goods, short product life cycles, seasonal demand shifts, and high price sensitivity add complexity to forecasting [51].

1. Price vs. Consumer composite sentiment index

In the Vietnamese F&B market, where price sensitivity is heightened by economic uncertainty, even small price adjustments can trigger disproportionate changes in consumer sentiment. For instance, a 5% price increase in milk tea led to a 15% decline in purchase volume during a mid-2023 promotional campaign [52]. This subsequently significantly

affects the accuracy of demand forecasting models. When prices increase, consumers are more likely to reassess their needs, explore alternative options, or postpone their purchases unless the product is deemed essential. Conversely, discount strategies tend to attract customers and drive short-term increases in demand.

In the highly competitive food and beverage (F&B) market, where price sensitivity is increasing, even minor price adjustments—whether upward or downward—can trigger abrupt shifts in demand. Forecasting models that rely solely on historical sales are, therefore, at risk of generating biased or inaccurate predictions if they fail to incorporate pricing variables and aggregate consumer sentiment indices as independent inputs. Ignoring this relationship and relying exclusively on past data prevents forecasting models from accurately capturing the market's responsiveness to changes in price and sentiment. As a result, businesses may misestimate demand, leading to inventory shortages or surpluses, which directly undermines the efficiency of supply chain operations.

2. Advertisement vs. Consumer composite sentiment index

Advertising influences consumer perceptions, shaping demand by increasing brand awareness and purchase intent [53]. Promotions often generate temporary sales spikes, making it essential for forecasting models to factor in advertising timing and intensity. If businesses fail to account for promotional campaigns, they risk underestimating demand, leading to stockouts, or overestimating post-campaign sales, causing excess inventory. Integrating advertising expenditure and consumer response data can improve forecast reliability.

3. Events, Price, Promotion vs. Consumer composite sentiment index

The interaction between special events, price adjustments, and consumer sentiment is especially critical in Vietnam's event-driven consumption cycles. For instance, during International Women's Day (March 8), ShopeeFood reported a 37% spike in F&B order volume, driven by bundled promotions and discount codes [54]. During these periods, consumers are more likely to increase their spending on dining, gatherings, celebrations, and gifts, leading to a sharp, short-term surge in demand for food and beverage (F&B) products.

To capitalize on these opportunities, F&B businesses frequently launch promotional campaigns, offer seasonal combos, or introduce limited-edition products tailored to specific events. Such marketing efforts not only attract target segments such as female customers during Women's Day but also significantly boost overall consumer spending. According to [54], on March 8, many companies offered promotions, including complimentary gifts and discounts of up to 50%, resulting in a noticeable increase in customer traffic and requiring staff to operate at full capacity to meet the demand. If businesses fail to accurately encode these events into their forecasting data, models may misinterpret demand trends. This could lead to underestimation during high-demand periods, resulting in stockouts, or overestimation once the event has passed, causing excess inventory. Accurately incorporating these time-sensitive factors is, therefore, essential for improving forecast reliability and operational efficiency.

4. Government policy vs. Macroeconomic

Government policies, along with macroeconomic indicators such as inflation, GDP growth, unemployment rate, per capita income, or minimum wage, play a crucial role in shaping the operating environment of the food and beverage (F&B) industry. These factors, although indirect, have a profound impact on consumer behavior and the ability of businesses to operate and set prices, thereby clearly influencing the outcomes of demand forecasting in the food and beverage (F&B) sector.

When the government introduces new regulations on food safety, traceability, or tightens input quality controls, businesses are required to adjust their supply chains, often resulting in increased operational costs. These changes can affect pricing strategies and alter consumers' willingness to pay. Moreover, shifts in real income due to macroeconomic fluctuations or regulatory policies, such as increases in fuel prices, electricity tariffs, or value-added tax (VAT), can directly influence how frequently and how much consumers spend on F&B products. During periods of economic instability or high inflation, consumers tend to shift toward lower-priced alternatives or reduce the frequency of dining out. Such changes reflect a defensive demand pattern that cannot be fully captured by endogenous forecasting models relying solely on historical sales data. Therefore, in a market environment characterized by volatility due to policy and macroeconomic influences, forecasting models need to be "extended" to not only learn from past data but also to "interpret" and incorporate real-world economic dynamics.

A typical example is when the government imposes or raises excise taxes on sugary beverages to curb consumption and promote public health. This policy leads to higher product prices, which in turn influence consumer behavior and can result in a decrease in demand for sugary drinks. In response, F&B businesses must adjust their revenue and volume forecasting models accordingly and consider developing alternative products, such as sugar-free or low-sugar beverages, to meet emerging market demands [51].

5. Inventory level, E-commerce vs. Promotion

The relationship between inventory management and e-commerce in the food and beverage industry is characterized by their mutual dependence on ensuring customer satisfaction and optimizing operational efficiency. As F&B businesses expand their online sales channels, such as food delivery apps, websites, or e-commerce platforms, the pressure to maintain accurate, flexible, and synchronized inventory becomes more critical than ever.

From a forecasting perspective, the rise in online orders introduces greater volatility compared to traditional sales channels, especially for trending products or during flash sales campaigns. Conventional forecasting models often struggle to keep pace without incorporating input variables that reflect the frequency of online orders, user engagement on digital platforms, or product impressions over time. To enhance inventory forecasting accuracy, businesses should integrate order data from e-commerce channels into their forecasting models, enabling them to adjust procurement plans in real-time and respond promptly to fluctuating market demand.

The online food delivery market in Vietnam reached a value of VND 52.4 trillion in 2023, marking a 20.18% increase compared to 2022 and a 3.5-fold rise from the pre-

pandemic period. This growth reflects the increasing popularity of online food ordering among Vietnamese consumers, driven by its convenience and ease of access. It also represents fertile ground that F&B businesses can effectively tap into alongside traditional on-site sales. Among F&B businesses using online food delivery platforms, 47.9% reported that online sales account for 25–50% of their total revenue. [55]. Beyond its growing scale, the market also demonstrates stable consumer habits and clear competition among food delivery platforms. The frequency of online food orders among Vietnamese consumers remains high, with 29.4% placing orders once or twice a week and 20% ordering three to four times a week. In terms of market share, 2023 saw significant expansion by ShopeeFood, which became the most-used platform among F&B businesses at 42.94%, surpassing GrabFood by 2.33% [55]. The strong growth in online food delivery revenue underscores the increasing importance of this channel within the modern food and beverage (F&B) industry landscape. Therefore, accurately forecasting demand across e-commerce channels is essential for optimizing production and distribution, maintaining adequate inventory levels, ensuring timely deliveries, and preventing stockouts.

6. Physical environment, Consumer composite sentiment index vs. E-commerce

The physical environment is one of the external factors that can significantly and directly influence the consumer sentiment index in the food and beverage (F&B) industry. Unlike internal factors such as pricing or promotional programs, which businesses can control, the physical environment is uncertain but can still be anticipated through meteorological data. This offers a significant advantage in improving forecast accuracy. For example, on hot days, the demand for cooling beverages such as milk tea, smoothies, and fruit juices tends to rise sharply, while demand for hot dishes typically decreases. In contrast, when the weather turns cold or during extended periods of rain, consumers often prefer hot food and drinks. They are less likely to go out, resulting in a higher number of delivery orders [56]. These changes can occur daily or even hourly, and if forecasting models fail to reflect the influence of the physical environment, the results can be significantly inaccurate.

In addition to affecting total demand, the physical environment also influences the form of purchase, whether it is direct or through delivery. On sunny or rainy days, consumers tend to stay indoors and place more orders through e-commerce platforms. This means F&B businesses must be able to accurately forecast not only the types of products that will be in demand but also the expected share of online orders based on actual weather conditions.

Therefore, it is necessary to integrate the method in forecasting Vietnam's F&B industry, which has the interrelationship between internal and external variables, each interacting in nonlinear ways. Relying solely on historical sales is inadequate in such a dynamic perspective. This paper highlights the need for a forecasting approach that integrates structured (e.g., price and inventory level) and unstructured (e.g., event, promotion, and sentiment) data.

B. Example of revenue forecasting

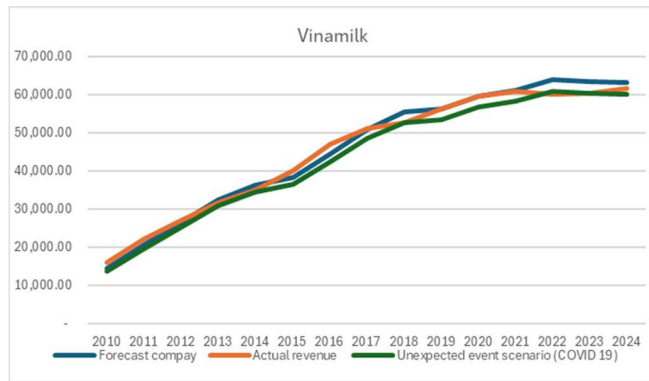


Figure 3. Forecasted vs. Actual Revenue of Vinamilk Under Normal and COVID-19 Scenarios (2010–2024)

Figure 3 illustrates the revenue trends of Vinamilk from 2010 to 2024 under three scenarios: the company's forecast, an unexpected event scenario (COVID-19), and the actual revenue. The chart shows that during the period 2020–2022, the company's forecast slightly overestimated revenue because it did not account for external shocks such as the COVID-19 pandemic.

The model integrated COVID-19 factor enables a more accurate representation of the decline in consumer demand and supply chain disruptions during the pandemic period. The adjusted scenario aligns closely with the actual revenue data, effectively capturing both the downturn during the pandemic and the subsequent gradual recovery.

This example highlights the importance of adopting an integrated forecasting approach that incorporates both internal and external factors. External shocks, such as COVID-19, significantly enhances the model's predictive capability, enabling businesses to align their strategies more effectively with real-world market dynamics. In contrast, relying solely on historical sales trends without considering external influences can lead to substantial forecasting errors and operational inefficiencies.

The factors identified in this study can serve as a base dataset for evaluating key variables that influence revenue forecasts. Furthermore, the proposed method can be utilized as a practical approach to improve forecasting accuracy.

V. CONCLUSIONS

The F&B industry operates in a highly dynamic environment where multiple internal and external factors interact to shape demand patterns. Internal drivers, such as pricing, promotions, advertising, inventory management, and e-commerce performance, combined with external influences including macroeconomic conditions, government policies, consumer sentiment, and disruptive events like the COVID-19 pandemic. These interactions create dynamic forecasting which can reflect real-world market behavior.

This study presents the critical importance of integrating both internal and external factors into revenue forecasting models, particularly in the context of disruptive events such as the COVID-19 pandemic. The findings demonstrate that reliance on historical data alone is insufficient for effective forecasting in dynamic environments. The proposed integrated approach not only enhances predictive accuracy but also provides a foundation for strategic decision-making. The identified factors can serve as a basis for further analysis and

supporting businesses in developing more resilient and responsive forecasting systems.

The proposed forecasting method remains under development, with further analyses, such as Exploratory Factor Analysis (EFA) and ARIMA modeling, required to identify and validate the most significant factors affecting F&B revenue in Vietnam. Nonetheless, this study serves as a foundational step by outlining key internal and external variables that potentially impact revenue forecasts. The integrated approach proposed here can also be applied in future studies to refine factor selection and enhance model accuracy.

In conclusion, F&B companies are encouraged to shift from sales-driven forecasting systems to adaptive, data-driven models that integrate both internal and external factors. The integrated systems enable businesses to navigate uncertainty, maintain competitiveness, and build operational resilience in an increasingly volatile and fast-changing marketplace.

ACKNOWLEDGMENT

The first author gratefully acknowledges the support received from the Excellent Foreign Students (EFS) scholarship, awarded by Sirindhorn International Institute of Technology (SIIT), Thammasat University, for this project.

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